CPE007
Programming Logic and Design

Final Project

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Ginhawa Clinic Record System

I. INTRODUCTION

The Ginhawa Clinic Record System is a C++ program developed to simplify and organize the management of clinic records. Its main purpose is to provide a centralized system where clinic staff can store, update, and retrieve patient information more conveniently. The program also allows users to view doctors and their respective specializations, as well as check their available schedules when managing clinic services. By organizing important information in one system, the program helps reduce the need for relying entirely on manual record keeping, which can take time and make it more difficult to locate specific information. The system is designed to make common clinic tasks easier to manage and help ensure that records are organized and accessible when needed.

The system also includes functions for managing patient appointments and recording consultation details, including diagnoses and medical history. These features help clinic staff keep track of patient visits and maintain important information related to previous consultations. Having these records organized in one program can make it easier to review patient information, monitor appointment schedules, and manage the daily activities of the clinic. In addition, the system aims to minimize errors that may occur when handling records manually and reduce the possibility of information being misplaced or overlooked. Overall, the Ginhawa Clinic Record System aims to improve the efficiency and organization of clinic record management by combining essential functions into one C++ program. Through this project, the developers aim to demonstrate how programming can be used to address practical problems

and provide a more convenient way of managing patient information, doctor schedules, appointments, and consultation records.

II. STATEMENT OF THE PROBLEM

Many clinics still rely on manual record- keeping, which can lead to misplaced patient records, inaccurate data information and delay in retrieving important data. These issues make it difficult for clinic staff to organize, update and manage patients' information efficiently. To address this issue, This study aims to develop a computerized Clinic record system using C++ to help the clinic stuff improve the management, organization and accessibility of patients records especially it seek answers to following questions:

1. How can a computerized Clinic records System in C++ help improve the storage and management of the patients records?
2. How can the system make it easier for clinic staff to add, update, search and manage patients' information?
3. How can the proposed system improve the efficiency of the clinic's record management and the quality of its services?

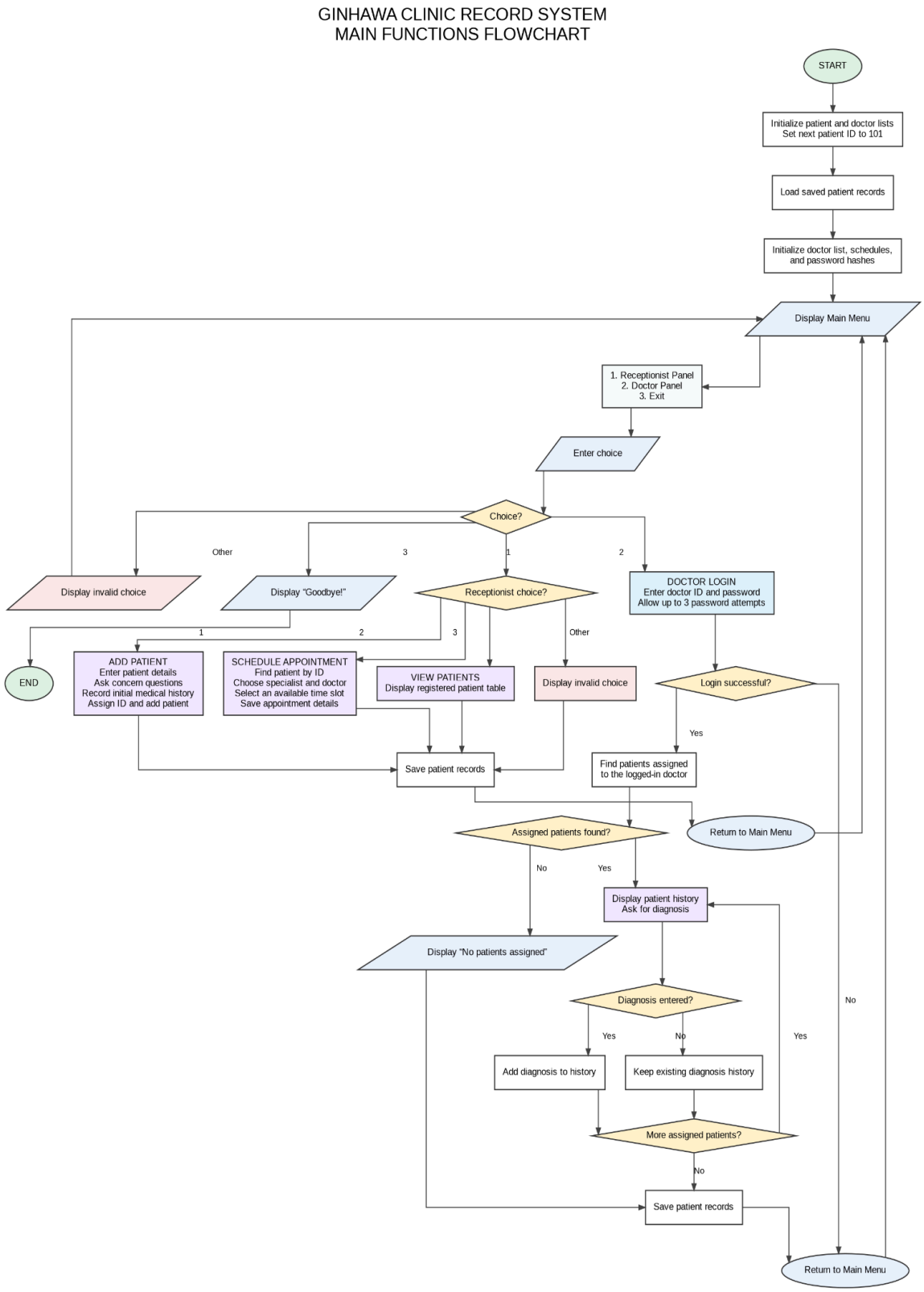
III. OBJECTIVES

The system aims to:

1. Develop a computerized platform that can store, organize, and easily recover patient information and medical history efficiently.
2. Develop a system that manages patient appointments and assigns available doctors based on their specialization.
3. Implement a record management module that allows staff to input, update, and search consultation details and diagnoses for improved accessibility and accuracy.

4. Enhance clinic operations by reducing manual workload and improving the reliability of patient data handling.

IV. FLOWCHART OF THE SYSTEM



V. PSEUDOCODE

```
BEGIN
LOAD patient records from file
SET nextPatientID = 101
INITIALIZE doctor list
FOR each doctor
        Generate salt
        Hash doctor password
END FOR
REPEAT
        DISPLAY "GINHAWA CLINIC RECORD SYSTEM"
        DISPLAY "1. Receptionist Panel"
        DISPLAY "2. Doctor Panel"
        DISPLAY "3. Exit"
        INPUT mainChoice
        IF mainChoice = 1 THEN
        DISPLAY "1. Add Patient"
        DISPLAY "2. Schedule Appointment"
        DISPLAY "3. View Patients"
        INPUT receptionistChoice
        IF receptionistChoice = 1 THEN
        INPUT patient name
        INPUT blood type
        INPUT sex
        INPUT address
        INPUT contact number
```

```
ASK health-related questions
DETERMINE patient concern
INPUT initial medical history
CREATE patient record
ASSIGN patient ID
SAVE patient record
DISPLAY success message
ELSE IF receptionistChoice = 2 THEN
DISPLAY all patients
INPUT patient ID
IF patient exists THEN
        DISPLAY patient concern
        DISPLAY specialist options
        INPUT specialist choice
        DISPLAY doctors under chosen specialty
        INPUT doctor selection
        DISPLAY doctor's available time slots
        INPUT slot ID
        IF slot is available THEN
        MARK slot as booked
        SCHEDULE appointment with:
        Date
        Time
        Doctor Name
        Specialty
        DISPLAY appointment scheduled successfully
```

```
        ELSE
        DISPLAY "Slot unavailable"
        END IF
ELSE
        DISPLAY "Patient not found"
END IF
ELSE IF receptionistChoice = 3 THEN
DISPLAY patient table
END IF
SAVE all patient records
ELSE IF mainChoice = 2 THEN
DISPLAY doctor list
INPUT doctor ID
IF doctor ID is valid THEN
SET attempts = 3
REPEAT UNTIL login successful OR attempts = 0
        INPUT password
        HASH entered password
        IF hash matches stored hash THEN
        DISPLAY doctor information
        FOR each patient
        IF patient is assigned to doctor THEN
                DISPLAY patient history
                INPUT diagnosis
                IF diagnosis not empty THEN
                ADD diagnosis to patient history
```

```
                    END IF
                END IF
                END FOR
                SAVE patient records
                EXIT login loop
                ELSE
                attempts = attempts - 1
                DISPLAY remaining attempts
                END IF
            END REPEAT
            IF attempts = 0 THEN
                DISPLAY "Too many failed attempts"
            END IF
            ELSE
            DISPLAY "Invalid Doctor ID"
            END IF
            ELSE IF mainChoice = 3 THEN
            DISPLAY "Goodbye"
            EXIT PROGRAM
            ELSE
            DISPLAY "Invalid Choice"
            END IF
UNTIL mainChoice = 3
END
```

VI. DATA DICTIONARY

Data Name	Type	Description
slotId	Integer	Unique identifier for each time slot.
date	String	Date of appointment or schedule.
time	String	Time of appointment or schedule.
isAvailable	Boolean	Indicates if a time slot is free or booked.
id	Integer	Unique identifier for doctor or patient.
name	String	Full name of doctor or patient.
specialty	String	Doctor’s area of specialization.
calendarSlots	List of TimeSlot	Collection of available time slots for each doctor.
bloodType	String	Patient’s blood type.

sex	String	Patient's gender.
diagnosis	String	Medical diagnosis assigned by doctor.
patientConcern	String	Patient's reason for consultation or concern.
appointment	Appointment	Stores appointment details.
hasAppointment	Boolean	Indicates if a patient already has an appointment.
nextId	Integer	Tracks the next available patient ID number.
choice	Integer	User's menu selection.
patients.txt	File	Stores saved patient data.
doctors.txt	File	Stores saved doctor data.

VII. SYSTEM SPECIFICATION

Item	Specification
System name	Ginhawa Clinic Record System
Programming language	C++
Operating System	Windows 11

VIII. HARDWARE SPECIFICATION

Device	ASUS TUF A15
CPU	AMD Ryzen 5 4600H
GPU	NVIDIA GeForce GTX 1650 Mobile
Processor	Up to 8 cores / 16 threads, boost clocks up to 5.2 GHz
RAM	24gb DDR4 3200mhz
Connectivity	Wi-Fi 6/6E, Bluetooth 5.2
Storage	512 GB PCIe 3.0/4.0 SSD standard
Operating System	Windows 11